Appendix 1: Assignment submissioncover sheet

**ASSIGNMENTSUBMISSIONCOVER SHEET**

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**Assignment title:** OOAD

**Date of submission:** 29/10/2026

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**Programme of Study:** CSE

**Year of Study:** YEAR 2

**Intellectual property statement**

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**Signature**.....L Mogale.....

**Date**.....29/10/26.....

# Part A – System Documentation Instructions

## 1.REQUIREMENT ELICITATION

i. Functional Requirements:

**-Customer Registration**

The system shall allow new customers to register by entering their personal details, such as their name and contact information. It should also allow the system to store these details so that customers can be identified and linked to their bank accounts.

- User Authentication

The system shall allow registered customers and the bank administrator to log in using their credentials. It shall verify their details before granting access and ensure that users can only access functions they are authorized to use.

- Account Management

The system shall allow customers to open and view their bank accounts. It shall support Savings, Investment, and Cheque Accounts, each with its own rules. For example, Savings

Accounts shall not allow withdrawals, while Investment Accounts shall earn monthly interest of 5% and Savings Accounts shall earn monthly interest of 0.05%.

-Deposits and Withdrawals

The system shall allow customers to deposit money into their accounts and withdraw money from eligible accounts. It shall update account balances after each transaction and prevent withdrawals from Savings Accounts.

- Interest Calculation
  The system shall automatically calculate and apply the correct monthly interest to Savings and Investment Accounts. The system shall update the account  balances after interest has been added.

-Transaction History

The system shall record all transactions, including deposits, withdrawals, and interest added to accounts. Customers shall be able to

view their transaction history to track their banking activities.

-Customer and Account Details

The system shall allow customers to view their account details and current balances. It shall also allow the bank administrator to view customer information and monitor accounts and transactions.

ii. Non- functional Requirements

-Security

The system shall protect customer and banking information from unauthorized access.

-Authentication

The system shall verify users' identities before granting access to banking functions.

-Accuracy

The system shall calculate account balances and interest correctly.

-Reliability

The system shall operate consistently and process transactions correctly.

-Usability

The system shall have a simple, clear, and user-friendly interface.

-Performance

The system shall process deposits, withdrawals, and balance enquiries within a reasonable time.

- Data Integrity

The system shall maintain accurate and consistent customer, account, and transaction records.

- Maintainability

The system shall be easy to update, troubleshoot, and maintain.

- Privacy

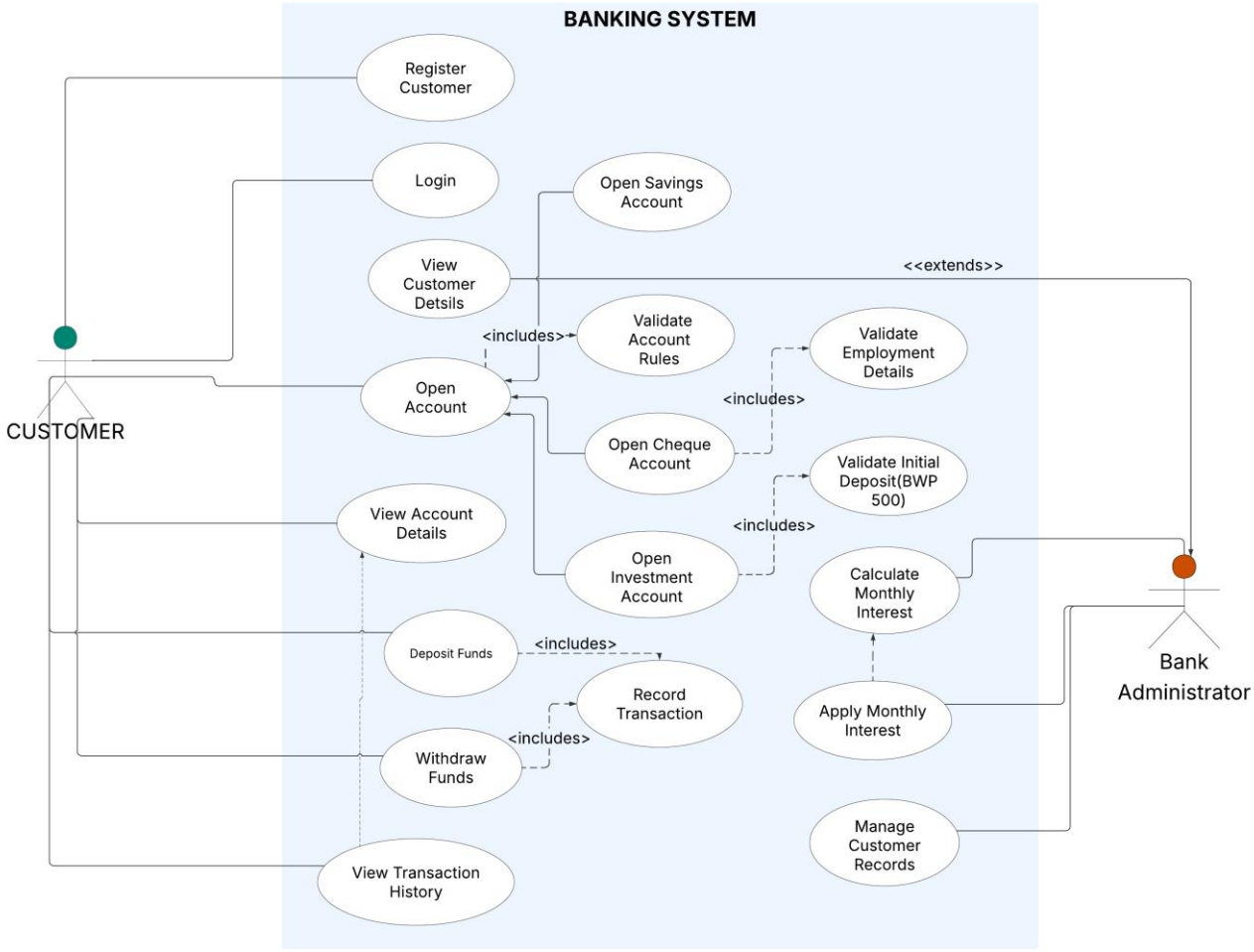
The system shall ensure that personal and financial information is accessible only to authorized users.

- Compatibility

The system shall be developed in Java and run in a compatible Java environment.

# 1. Structural UML Modelling

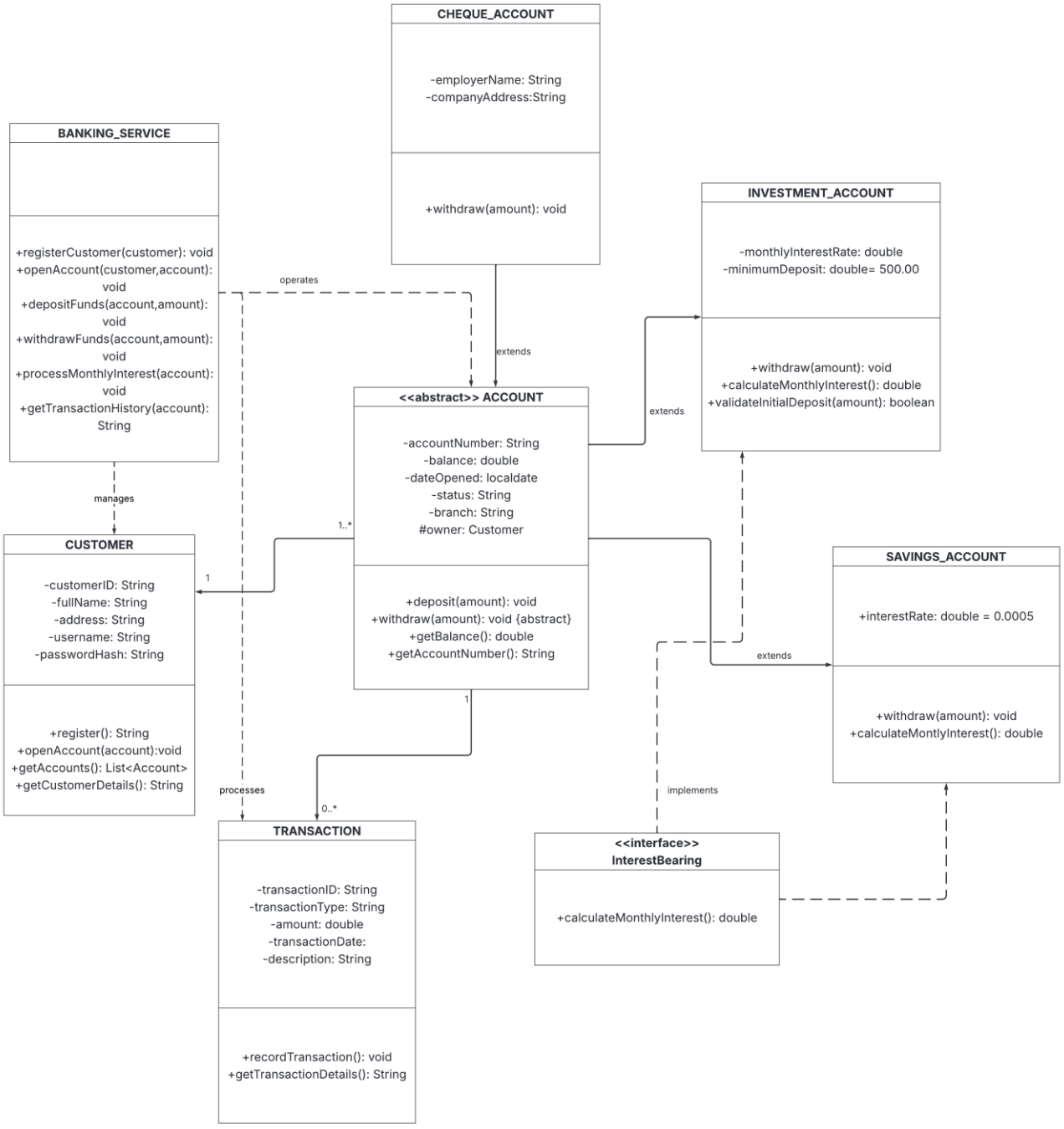
## System Use Case Diagram





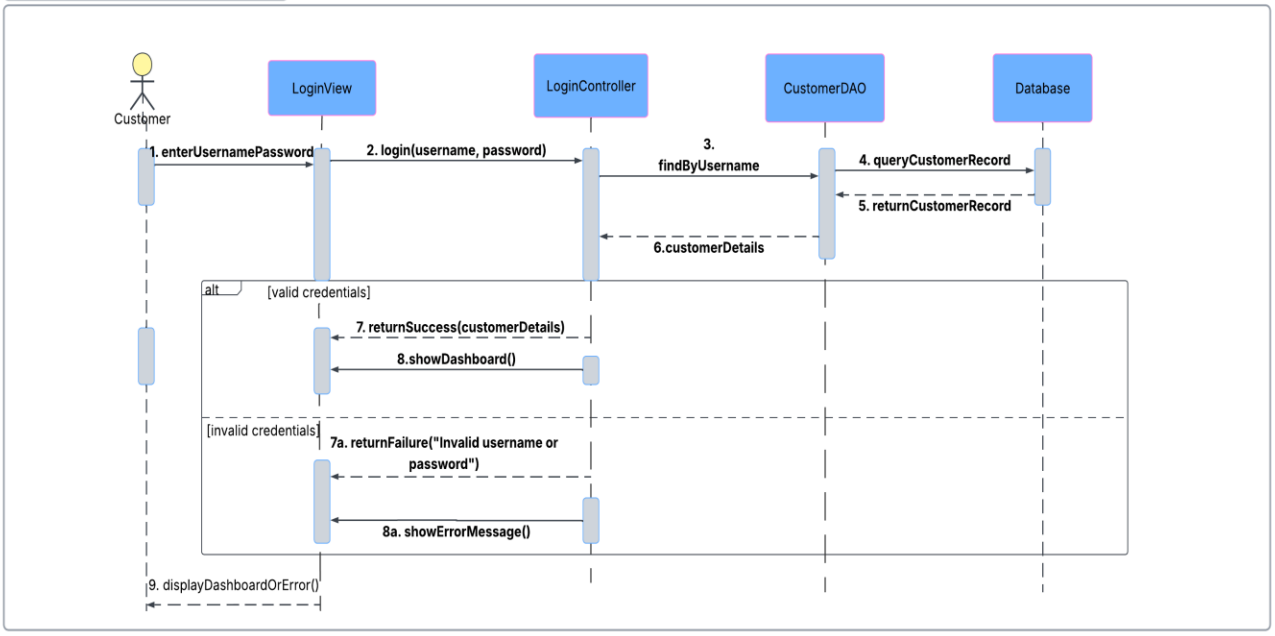

# Class Diagram

BANKING SYSTEM

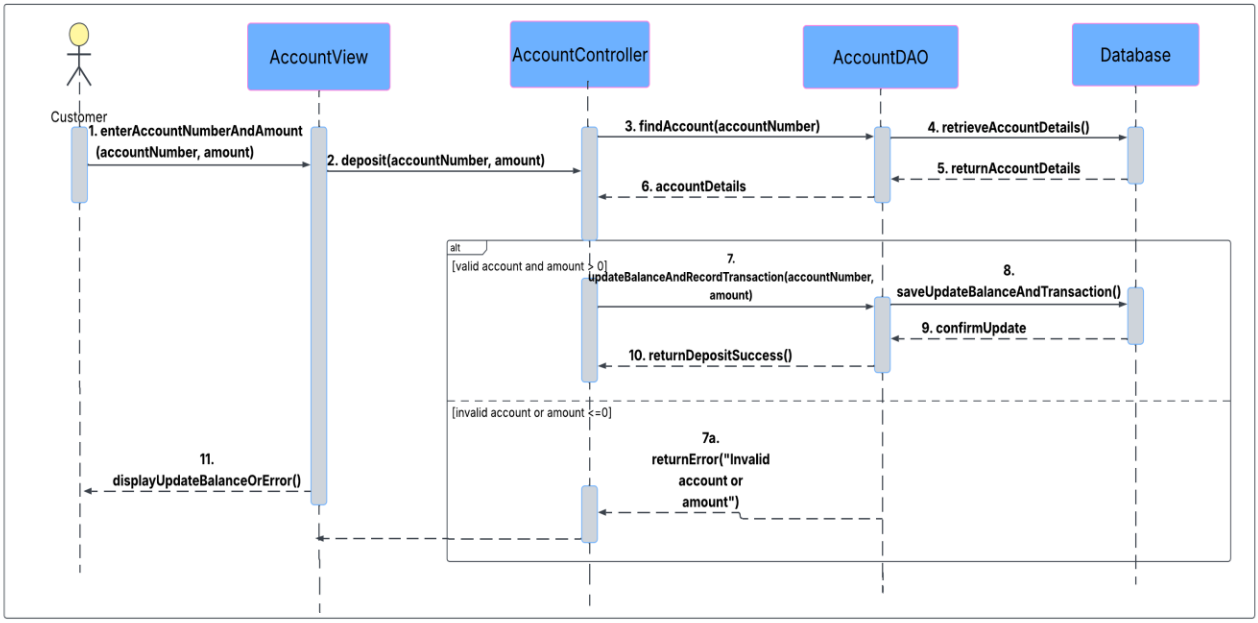


# Sequence Diagram

## 1. Login Sequence Diagram




## 2. Deposit Funds Sequence Diagram



# State Diagram

